PARTNERSHIP PLAN

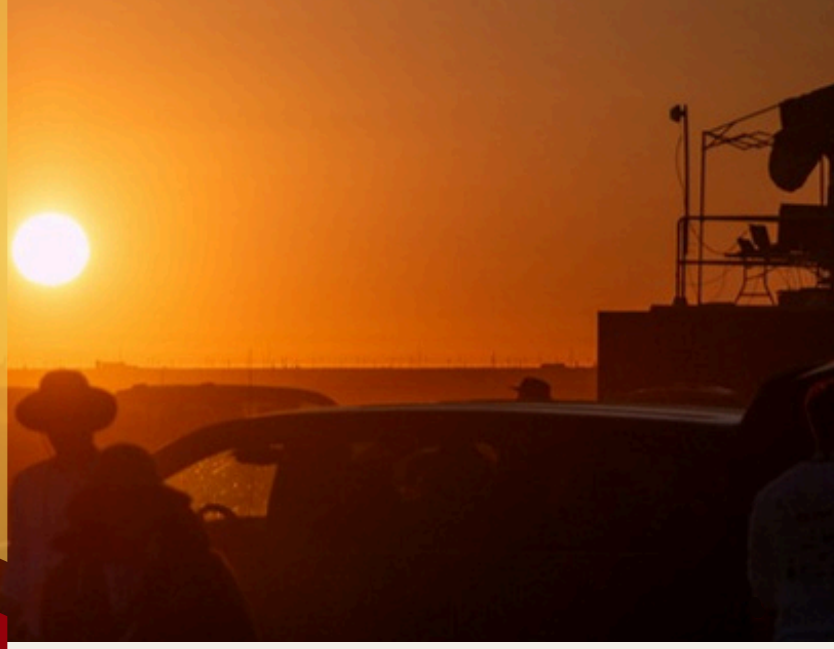
GAUL

GROUPE AÉROSPATIAL DE L'UNIVERSITÉ LAVAL

UNIVERSITÉ LAVAL

Conceptualized by Marguerite Ricard et Florence Boucher

Groupe aérospatial de l'université Laval



Our divisions

Aerostructure

- Design and manufacturing of the rocket's structural components, as well as the integration of the rocket's other systems.
- Materials used: composite materials such as fiberglass and carbon fiber.

Propulsion

- Development, design, and manufacturing of a liquid engine.
- Equipment and materials used: isopropanol, valves, nitrous oxide (N_2O), machining services, and instrumentation for measuring various parameters during testing

Payload

- Design and development of a payload integrated into the rocket, including the creation of a hyperspectral camera and a rover.
- Equipment used: 3D printer and 3D-printing plastics, electronic bays.

Avionics

- Design of electronic circuits for our rockets and for several of our ground-based electronic systems.
- Equipment used: PCBs, RF antennas, altimeters, onboard computers, GPS systems, telemetry, and arming systems.

Who we are

The Université Laval Aerospace Group (GAUL), founded in 1993, brings together around fifty students from the Faculty of Science and Engineering who work collaboratively on the design and construction of high-power rockets launched in both national and international competitions.

The many projects at GAUL provide our members with the opportunity to apply the knowledge gained throughout their academic journey while developing essential engineering and research skills such as teamwork, problem-solving, and project management.

Contact Information

gaul@fsg.ulaval.ca

https://gaulspace.web.app/home

https://www.instagram.com/gaul.ul/

https://www.facebook.com/groupeaerospatialul

http://www.youtube.com/@projetGAUL

Conceptualized by Marguerite Ricard et Florence Boucher

COMPETITIONS

Launch Canada

The first national rocket competition in Canada, designed to inspire and foster aerospace engineering talent. This event brings together university students and rocketry enthusiasts to drive innovation and advance the future of Canadian aerospace.

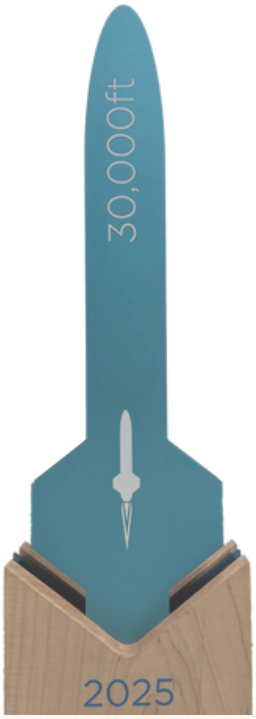
IREC

The International Rocket Engineering Competition (IREC) is an event where student rocketry teams from around the world compete by launching their rockets. The competition is open to student teams worldwide, including those from the United States.

Far-Out

The FAR-OUT (Friends of Amateur Rocketry: Oxidizers Uninhibited Tournament) is a student rocketry competition focused on hybrid and liquid-propellant engines, giving university teams the opportunity to launch their rockets to target altitudes while carrying a payload.

OUR ACHIEVEMENTS OF 2024-2025

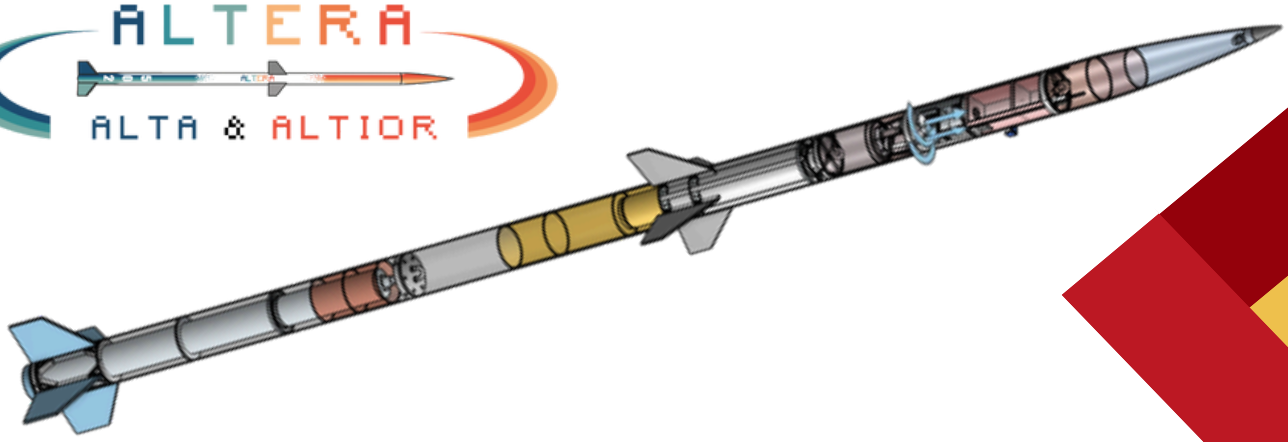


In 2025, GAUL took on a new challenge by designing and launching ALTERA, a two-stage rocket in the 30,000-foot category. It was the first two-stage rocket built by the current team. The project concluded successfully with a nominal flight and an altitude of 30,403 feet, earning GAUL first place in our category and 11th place overall across all categories, out of a total of 152 teams.

Our most recent solid-motor rockets launched at Launch Canada were MAIA in 2024, where we ranked 4th with an altitude of 14,477 feet, and PAGOMA in 2025, where we ranked 7th with 12,021 feet.

ALTERA

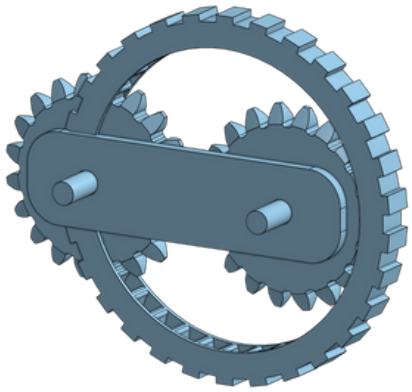
ALTA & ALTIOR



Conceptualisé par Marguerite Ricard et Florence Boucher

PROJECTS FOR 2025-2026

Our rover's first wheel prototype



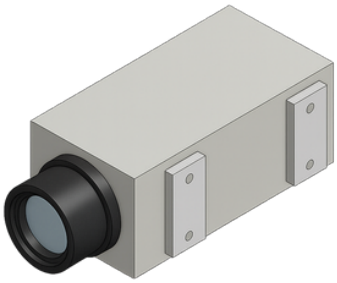
One of our ongoing projects is to design a two-stage rocket capable of reaching 100,000 feet. Achieving this milestone would allow us to break the Canadian record of 75,000 feet for an amateur rocket.

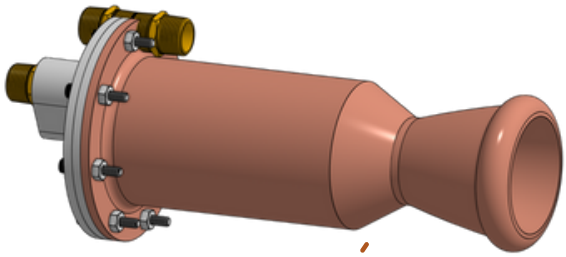
This rocket will include a payload equipped with a hyperspectral camera, intended to perform a chemical analysis of the atmosphere.

This year, we aim to build on ALTERA's remarkable success. We are preparing to launch a new two-stage rocket, this time in the highest category, targeting an altitude of 45,000 feet, which will be flown at the IREC.

In parallel, GAUL will participate in a secondary category at IREC showcasing one of our innovative projects. Our goal: to design a rover that will be deployed during the rocket's descent.

Preview of our hyperspectral camera model





Preliminary model of our liquid engine

Our third project of the year represents a new challenge for GAUL: it will be our first liquid-engine rocket. This single-stage vehicle will be launched in the Category B – Liquids at the FAR-OUT competition, which requires teams to reach an altitude between 20,000 and 50,000 feet.

JOIN US !

The completion of all these projects requires significant resources. It would therefore be impossible for us to carry them out without the support of our valued partners. By joining our team through financial or material contributions, you will directly support the pursuit of our ambitions. In return, we commit to offering you privileged visibility during our competitions as well as at the numerous events we participate in each year.

PARTNERSHIP CATEGORIES

Benefits	Terrestrial Orbit 499 $ and less	Interplanetary 500 $ to 2499 $	Interstellar 2500 $ to 4999 $	Intergalactic 5000 $ and more
Logo on our website	X	X	X	X
Logo on competition shirts	X	X	X	X
Invitation to networking events		X	X	X
Logo on our rockets		X	X	X
Logo on promotional event material		X	X	X
Written description of your company at national events			X	X
Acknowledgment on our social media			X	X
Logo on posts during competitions				X
Mention during presentations				X
Written description of your company at international events				X

Our 2025 sponsors

Intergalactic

KELLY MACHINERIE

AESGUL

ASDevices

Pratt & Whitney
Une société de RTX

AtkinsRéalis

Interstellar

Eddyfi Technologies

onshape

UNIVERSITÉ LAVAL

MAGICOLOR
PEINTURE INDUSTRIELLE

Interplanetary

La Fusée

THETFORD ARMATURE

TEXONIC

KAIZEN BUDO INTERNATIONAL

AtkinsRéalis

Les Entreprises MATRICK inc.

TeXtreme®

Terrestrial Orbit

Assek Technologies

TeraXion
An indie Semiconductor Company

WEST SYSTEM

AEGMUL

Victoria

RÉGIÉ

aegep

Smpr inc.

CADEUL

Conceptualisé par Marguerite Ricard et Florence Boucher